Musterlösungen Probetest 1 (Kap. 1-4) - koma - HS 2026

① a.) $3.5 \cdot 10^{-17} - 9.0 \cdot 10^{-18} - 40.0 \cdot 10^{-19}$

$= 350.0 \cdot 10^{-19} - 90.0 \cdot 10^{-19} - 40.0 \cdot 10^{-19}$

$= (350.0 - 90.0 - 40.0) \cdot 10^{-19} = 220.0 \cdot 10^{-19} = \underline{\underline{2.2 \cdot 10^{-17}}}$

b.) $4\sqrt{2} + \frac{3}{\sqrt{2}} - 2\sqrt{\frac{9}{2}} = 4\sqrt{2} + \frac{3\sqrt{2}}{\sqrt{2}^2} - 2\frac{\sqrt{9}}{\sqrt{2}}$

$= 4\sqrt{2} + \frac{3\sqrt{2}}{2} - \frac{2 \cdot 3 \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}} = \frac{8\sqrt{2} + 3\sqrt{2} - 6\sqrt{2}}{2} = \underline{\underline{\frac{5\sqrt{2}}{2}}}$

② a.) $(-2x^{-2})^3 \cdot (3x^{-2})^{-3} = \left(-\frac{2}{x^2}\right)^3 \cdot \frac{1}{(3/x^2)^3}$

$= (-1)^3 \frac{2^3}{(x^2)^3} \cdot \frac{\cdot 1}{3^3/(x^2)^3} = -\frac{8}{x^{2 \cdot 3}} \cdot \frac{(x^2)^3}{27} = -\frac{8x^6}{27x^6} = \underline{\underline{-\frac{8}{27}}}$

b.) $\left(\frac{a}{2} - \frac{2}{b}\right)^{-4} \cdot \left(\frac{b}{ab-4}\right)^{-3} = \left(\frac{ab-4}{2b}\right)^{-4} \cdot \left(\frac{ab-4}{b}\right)^3$

$= \left(\frac{2b}{ab-4}\right)^4 \cdot \frac{(ab-4)^3}{b^3} = \frac{2^4 \cdot b^4 (ab-4)^3}{(ab-4)^4 \cdot b^3} = 16 \cdot b^{4-3} \cdot (ab-4)^{3-4}$

$= 16b \cdot (ab-4)^{-1} = \underline{\underline{\frac{16b}{ab-4}}}$

③ a.) $\sqrt[6]{\frac{x \sqrt[4]{x^3}}{\sqrt[12]{x^3}}} = x^{1/6} \cdot x^{\frac{3}{6 \cdot 4}} \cdot x^{-\frac{3}{6 \cdot 12}}$

$= x^{\frac{1}{6} + \frac{1}{8} - \frac{1}{24}} = x^{\frac{4+3-1}{24}} = x^{6/24} = x^{1/4} = \underline{\underline{\sqrt[4]{x}}}$

b.) $\frac{a}{2 - \sqrt{a}} = \frac{a(2+\sqrt{a})}{(2-\sqrt{a})(2+\sqrt{a})} = \frac{2a + a\sqrt{a}}{2^2 - \sqrt{a}^2} = \underline{\underline{\frac{a(2+\sqrt{a})}{4-a}}}$

c.) $\frac{x^2}{2} \cdot \sqrt[3]{\frac{16}{x^5}} = \sqrt[3]{\left(\frac{x^2}{2}\right)^3} \cdot \sqrt[3]{\frac{16}{x^5}} = \sqrt[3]{\frac{x^6 \cdot 16}{8 \cdot x^5}} = \underline{\underline{\sqrt[3]{2x}}}$

d.) $\sqrt[4]{\frac{32x^5y}{3z^9}} = \sqrt[4]{\frac{16x^4}{z^8} \cdot \frac{2xy}{3z}} = \sqrt[4]{\left(\frac{2x}{z^2}\right)^4} \cdot \sqrt[4]{\frac{2xy}{3z}} = \underline{\underline{\frac{2x}{z^2} \sqrt[4]{\frac{2xy}{3z}}}}$

④ a.) • $\lg\left(\frac{1}{100000}\right) = \lg(10^{-5}) = \underline{\underline{-5}}$

• $\log_7(1) = \log_7(7^0) = \underline{\underline{0}}$

• $\log_5(125) = \log_5(5^3) = \underline{\underline{3}}$

• $\log_4(32) = \log_4(16 \cdot 2) = \log_4(16) + \log_4(2)$

$= \log_4(4^2) + \log_4(\sqrt{4}) = 2 + \frac{1}{2} = \underline{\underline{5/2}}$

b.) $2 + \frac{1}{3}\left[\log_3(a) - \frac{3}{4}\log_3(b) + 2\log_3(c)\right]$

$= 2 + \frac{1}{3}\log_3\left(\frac{ac^2}{\sqrt[4]{b^3}}\right) = \log_3(2^3) + \log_3\left(\sqrt[3]{\frac{ac^2}{\sqrt[4]{b^3}}}\right)$

$= \underline{\underline{\log_3\left(8 \cdot \sqrt[3]{\frac{ac^2}{\sqrt[4]{b^3}}}\right)}}$

c.) $\log_4\left(\frac{a^2-b^2}{4\sqrt{a+b}}\right) = \log_4\left(\frac{(a+b)(a-b)}{4\,(a+b)^{1/2}}\right) = \log_4\left(\frac{(a+b)^{1/2}(a-b)}{4}\right)$

$= \frac{1}{2}\log_4(a+b) + \log_4(a-b) - \log_4(4)$

$= \underline{\underline{\frac{1}{2}\log_4(a+b) + \log_4(a-b) - 1}}$

⑤ a.) $\sqrt{x+5} + 1 = x \quad / -1$

$\sqrt{x+5} = x-1 \quad / ()^2$ ⚠

$x+5 = x^2 - 2x + 1 \quad / -x-5$

$0 = x^2 - 3x - 4$

$0 = (x-4)(x+1)$

$\Rightarrow x_1 = 4$ ✓ $\quad x_2 = -1$ ✗ $\quad \Rightarrow \underline{\underline{\mathbb{L} = \{4\}}}$

b.) $\sqrt{x+1-\sqrt{x+3}} = 2 \quad / ()^2$ ⚠

$x+1-\sqrt{x+3} = 4 \quad / -4 + \sqrt{x+3}$

$x-3 = \sqrt{x+3} \quad / ()^2$ ⚠

$x^2 - 6x + 9 = x + 3 \quad / -x-3$

$x^2 - 7x + 6 = 0$

$(x-6)(x-1) = 0$

$\Rightarrow x_1 = 6$ ✓

$\Rightarrow x_2 = 1$ ✗

$\Rightarrow \underline{\underline{\mathbb{L} = \{6\}}}$

⑥ a.) $2^x = 80 - 3 \cdot 2^{x-1} \quad | + 3 \cdot 2^{x-1}$

$2^x + 3 \cdot 2^x \cdot 2^{-1} = 80 \quad | \cdot 2$

$2 \cdot 2^x + 3 \cdot 2^x = 160$

$5 \cdot 2^x = 160 \quad | : 5$

$2^x = 32$

$2^x = 2^5 \quad | \log_2()$

$x = 5$

$\Rightarrow \underline{\underline{\mathbb{L} = \{5\}}}$

e.) $\lg(2x+13) + \lg(x+10) = \lg(x+20) + 1 \quad \Rightarrow \underline{D = (-6.5, \infty)}$

$\lg((2x+13)(x+10)) = \lg((x+20) \cdot 10) \quad | \ 10^{()}$

$(2x+13)(x+10) = 10(x+20)$

$2x^2 + 33x + 130 = 10x + 200 \quad | -10x - 200$

$2x^2 + 23x - 70 = 0$

$x_{1,2} = \frac{-23 \pm \sqrt{23^2 - 4 \cdot 2 \cdot (-70)}}{2 \cdot 2} = \frac{-23 \pm 33}{4}$

$x_1 = \frac{10}{4} = 2.5$ ✓

$x_2 = -\frac{56}{4} = -14$ (F)

$\Rightarrow \underline{\underline{\mathbb{L} = \{2.5\}}}$

⑦ $y = A_0 e^{\lambda x}$

P(1,10): $10 = A_0 \cdot e^{\lambda} \quad$ (I)

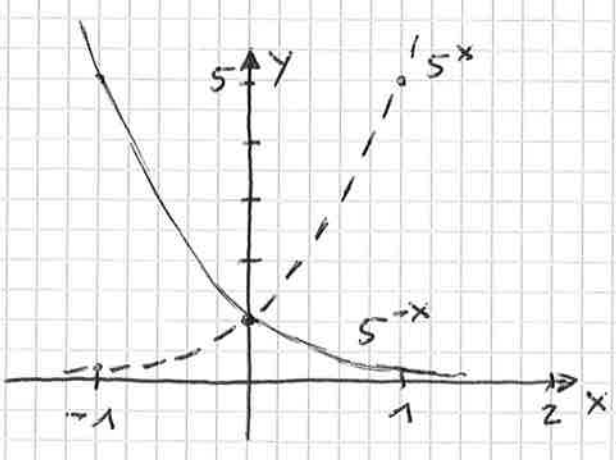
Q(4,5): $5 = A_0 \cdot e^{5\lambda} \quad$ (II)

$\frac{II}{I}$: $\frac{1}{2} = \frac{e^{5\lambda}}{e^{\lambda}} = e^{4\lambda} \quad \Rightarrow \lambda = \frac{1}{4} \ln(1/2) = \underline{-\frac{1}{4}\ln(2)}$

mit (I): $10 = A_0 \cdot e^{-\frac{1}{4}\ln(2)} = A_0 e^{\ln(2^{-1/4})} = A_0 \cdot 2^{-1/4}$

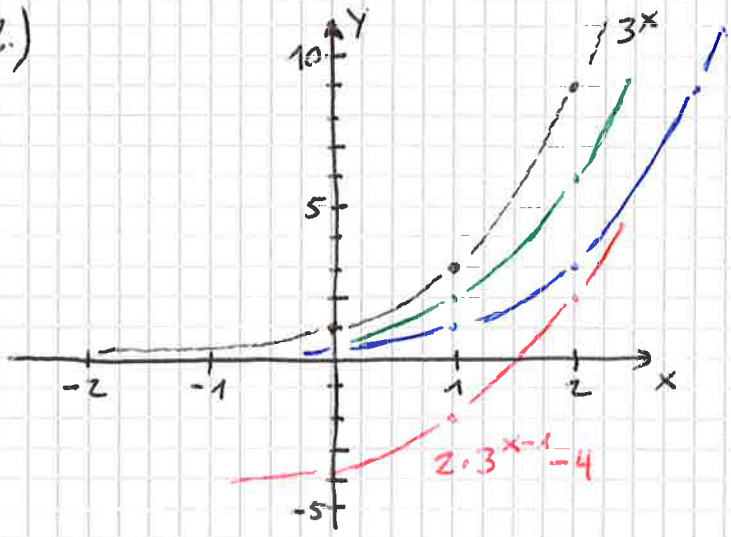
$\Rightarrow A_0 = 10 \cdot 2^{1/4} = \underline{\underline{10\sqrt[4]{2}}}$

⑧ a.)



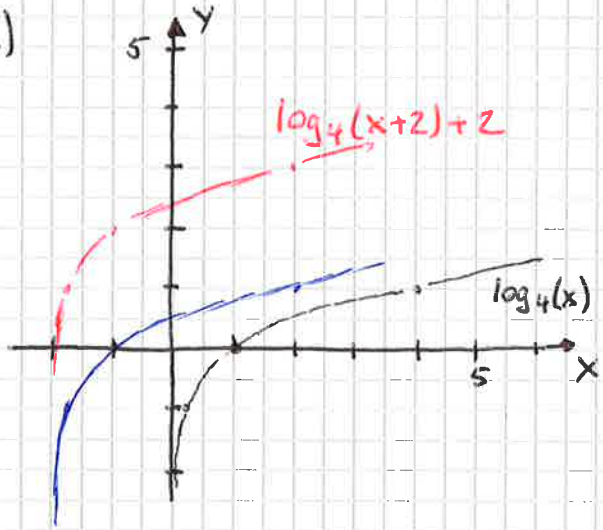
Transformation: Spiegelung an der y-Achse

b.)



Transformationen
- Verschiebung um 1 in x-Richtung
- Streckung um Faktor 2 in y-Richtung
- Verschiebung um -4 in y-Richtung

c.)



Transformationen
- Verschiebung um -2 in x Richtung
- Verschiebung um +2 in y-Richtung

(9) a.) $y = f(x) = -2x^2 + 12x - 15 = -2(x-3)^2 + 3$

$\Rightarrow$ Scheitelpunkt $x_S = -\frac{b}{2a} = -\frac{12}{2(-2)} = 3$

$y_S = -\frac{b^2-4ac}{4a} = 3$

$\Rightarrow$ bijektiv z.B. für $f: [3, \infty) \to (-\infty, 3]$

$\Rightarrow$ Umkehrfunktion

$y = -2(x-3)^2 + 3 \quad | -3$

$y-3 = -2(x-3)^2 \quad | : (-2)$

$\frac{3-y}{2} = (x-3)^2 \quad | \sqrt{\ }$

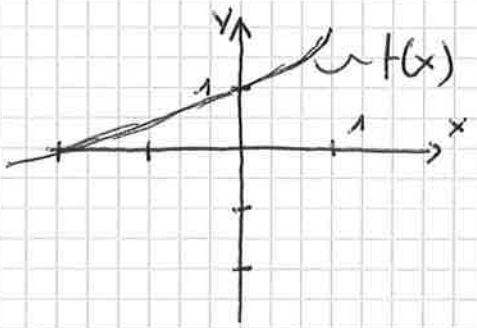
$\pm\sqrt{\frac{3-y}{2}} = x-3 \quad | +3$

$3 \pm \sqrt{\frac{3-y}{2}} = x$

$\Rightarrow f^{-1}: (-\infty, 3] \to [3, \infty)$

$x \mapsto y = f^{-1}(x) = 3 + \sqrt{\frac{3-x}{2}}$

e.) $y = f(x) = 2 - \sqrt{1-x}$ $\Rightarrow$ bijektiv für $f: (-\infty, 1] \to (-\infty, 2]$



$\Rightarrow$ Umkehrfkt.

$y = 2 - \sqrt{1-x}$ $\quad / -2$

$y - 2 = -\sqrt{1-x}$ $\quad / ()^2$

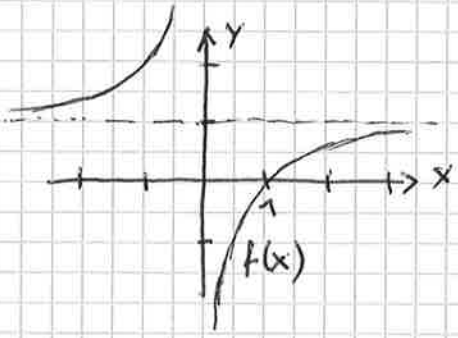
$(y-2)^2 = 1 - x$ $\quad / + x - (y-2)^2$

$x = 1 - (y-2)^2$

$\Rightarrow \quad f^{-1}: (-\infty, 2] \to (-\infty, 1]$

$x \mapsto y = f^{-1}(x) = 1 - (x-2)^2$

c.) $y = f(x) = \frac{x-1}{x} = 1 - \frac{1}{x}$ $\Rightarrow$ bijektiv für $f: \mathbb{R} \setminus \{0\} \to \mathbb{R} \setminus \{1\}$



$\Rightarrow$ Umkehrfkt.

$y = 1 - \frac{1}{x}$ $\quad / + \frac{1}{x} - y$

$\frac{1}{x} = 1 - y$ $\quad / ()^{-1}$

$x = \frac{1}{1-y}$

$\Rightarrow \quad f^{-1}: \mathbb{R} \setminus \{1\} \to \mathbb{R} \setminus \{0\}$

$x \mapsto y = f^{-1}(x) = \frac{1}{1-x}$

(10) a.) $K(t) = K_0 (1+p)^t$ $\Rightarrow$ $86400 = 50000 (1+p)^3$ $\quad / : 50000$

$\frac{864}{500} = (1+p)^3$ $\quad / \sqrt[3]{\ }$

$1.2 = 1 + p$ $\quad / -1$

$\Rightarrow \underline{p = 0.2 = 20\%}$

$\Rightarrow \underline{K(t) = 50000 \cdot 1.2^t}$

b.) $K(6) = 50000 \cdot 1.2^6 = \underline{149'299.20}$

c.) $K(t) = 2 \cdot 50000 = 50000 \cdot 1.2^t$ $\quad / : 50000$

$2 = 1.2^t$ $\quad / \log_{1.2}()$

$\Rightarrow t = \log_{1.2}(2) = \underline{3.8}$ (Jahre)

(11) a.) $B_E(t) = B_{E,0}\,(1+p)^t$

$\Rightarrow$ 2050 ergibt $t = 46$: $B_E(46) = 630 \cdot 10^6 = 725 \cdot 10^6 (1+p)^{46}$

$0{,}869 = (1+p)^{46} \quad / \sqrt[46]{\ }$

$0{,}997 = 1+p$

$\Rightarrow p = -0{,}003 = \underline{\underline{-0{,}3\,\%}}$

b.) $B_A(t) = B_{A,0} \cdot e^{\lambda t}$

$\Rightarrow 2050 \Rightarrow t = 46$: $1{,}8 \cdot 10^9 = 900 \cdot 10^6 \cdot e^{46\lambda}$

$2 = e^{46\lambda} \quad / \ln()$

$\ln(2) = 46\lambda$

$\Rightarrow \lambda = \frac{1}{46}\ln(2) = \underline{\underline{0{,}0151}}$

$\Rightarrow B_{A,0} = B_0 = 900 \cdot 10^6$ (Einwohner Afrika im Jahr 2004)

$\lambda = 0{,}0151$ (Wachstumskonst.)

c.) $B_A(t) = 900 \cdot 10^6 \cdot e^{0{,}0151 \cdot t} = 1{,}2 \cdot 10^9 \quad / : 900 \cdot 10^6$

$e^{0{,}0151 \cdot t} = 4/3 \quad / \ln()$

$0{,}0151 \cdot t = \ln(4/3) \quad / : 0{,}0151$

$\Rightarrow t = \frac{\ln(4/3)}{0{,}0151} = \underline{\underline{19{,}1}}$ (Jahre ab 2004)

$\Rightarrow$ Im Jahr $\underline{2023}$

d.) $2 \cdot 725 \cdot 0{,}997^t = 900 \cdot e^{0{,}0151 t} \quad / : 900$

$1{,}6111 \cdot 0{,}997^t = \left(e^{0{,}0151}\right)^t \quad / : 0{,}997^t$

$1{,}6111 = \frac{1{,}0152^t}{0{,}997^t}$

$1{,}6111 = 1{,}0183^t \quad / \log_{1{,}0183}()$

$\Rightarrow t = \log_{1{,}0183}(1{,}6111) = \underline{\underline{26{,}3}}$ (Jahre)

$\Rightarrow$ Im Jahr $\underline{2030}$